

Solution STA510 Exam fall 2025

Problem 1

We like to model and analyze the rainfall in Stavanger during a storm. The English term storm describes here a low-pressure area with some rain, and a storm event is defined hydrologically as a continuous precipitation episode separated from others by ≥ 6 hours of no rain. Let X be a random variable describing the *rainfall depth per storm* (mm per m^2). We assume that the rainfall depths for individual storms are independent and identically distributed as

$$X \sim \text{Exp}(\beta) \quad \text{with} \quad f_X(x) = \frac{1}{\beta} e^{-x/\beta}, \quad x \geq 0.$$

Assume further, we have collected $n = 40$ storm depths over a recent period:

2.5	4.0	6.0	7.0	8.0	9.0	10.0	11.0	12.0	13.0
5.0	13.0	7.0	14.0	9.0	11.0	8.0	10.0	6.0	12.0
3.0	5.0	9.0	10.0	8.0	7.0	15.0	14.0	9.0	12.0
6.0	13.0	11.0	8.0	10.0	9.0	16.0	5.0	7.0	17.0

with sample mean of $\bar{X} = 9.29mm$ and sample standard deviation of $\sigma = 3.53mm$.

a)

Assuming a scale parameter of $\beta = 8mm$ in Stavanger. i) Find the probability that it rains more than $12mm$ during a single storm event in Stavanger. ii) Given that it has already rained $5mm$ in a storm, find the probability that the storm's total rainfall exceeds $12mm$. \

i)

$$P(X > 12) = \int_{12}^{\infty} \frac{1}{8} e^{-x/8} = e^{-12/8} \approx 0.2231.$$

ii) We can either calculate

$$P(X > 12 | X \geq 5) = \frac{P(X > 12)}{P(X \geq 5)} = \frac{e^{-12/8}}{e^{-5/8}} \approx 0.4169.$$

or use the memoryless property

$$P(X > 12 | X \geq 5) = P(X > 7) = e^{-7/8} \approx 0.4169.$$

b)

Now, we like to estimate β from the data. i) Show the the Likelihood function is given by $L(\beta) = \beta^{-n} e^{-\frac{1}{\beta} \sum_{i=1}^n x_i}$. ii) Show that the Maximum Likelihood estimator (MLE) is $\hat{\beta} = \frac{1}{n} \sum_{i=1}^n x_i$. iii) Is $\hat{\beta}$ an unbiased estimator? Reason your answer! iv) Show that the variance of $\hat{\beta}$ is $\frac{\beta^2}{n}$.

i) The Likelihood function is

$$L(\beta) = \prod_{i=1}^n f(x_i; \beta) = \prod_{i=1}^n \beta^{-1} e^{-\frac{1}{\beta} x_i} = \beta^{-n} e^{-\frac{1}{\beta} \sum_{i=1}^n x_i}.$$

ii) The log-likelihood is

$$l(\beta) = -n \ln(\beta) - \frac{1}{\beta} \sum_{i=1}^n x_i.$$

Further

$$l'(\beta) = -\frac{n}{\beta} + \frac{1}{\beta^2} \sum_{i=1}^n x_i \Leftrightarrow n = \frac{1}{\beta} \sum_{i=1}^n x_i$$

and thus

$$\hat{\beta} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{X}$$

.

Further for the 2nd derivative we have

$$l''(\beta) = \frac{n}{\beta^2} - \frac{2}{\beta^3} \sum_{i=1}^n x_i = \frac{1}{\beta^2} (n - 2n) = -\frac{n}{\beta^2} < 0.$$

iii) We know $E(\bar{X}) = \frac{1}{n} \sum E(X) = \beta$ and thus $\hat{\beta}$ is an unbiased estimator.

iv) Further, $Var(\bar{X}) = \frac{1}{n^2} \sum Var(X) = \frac{\beta^2}{n}$.

c)

How many measurements of the rain depth should we take to be 95% sure that $\hat{\beta}$ falls at most a margin of 1mm from β ? Reason your answer! \

We see that the MLE for β is the sample average, thus we can use the results about the uncertainty of the mean estimate from the lectures. For, 95% probability, that $\hat{\beta}$ falls at most a margin of $e = 1mm$ from β away, we have to have $n > \frac{4\sigma^2}{e^2} = 4 \cdot 12.46 \approx 50$.

d)

Now, we want to draw $B = 4000$ non-parametric bootstrap samples from the 40 rainfalls depth above. Given the R-code and output below: i) Estimate the bias and the MSE for $\hat{\beta}$, and calculate the bias corrected estimate. ii) Construct a 95% standard normal bootstrap interval, 95% basic bootstrap interval, and 95% percentile bootstrap interval.

i)

```
rain <- c(2.5,4.0,6.0,7.0,8.0,9.0,10.0,11.0,12.0,13.0,
         5.0,13.0,7.0,14.0,9.0,11.0,8.0,10.0,6.0,12.0,
         3.0, 5.0, 9.0, 10.0, 8.0, 7.0, 15.0, 14.0, 9.0, 12.0,
         6.0, 13.0, 11.0, 8.0, 10.0, 9.0, 16.0, 5.0, 7.0, 17.0 )
n <- length(rain)

B <- 4000
estB <- numeric(B)
set.seed(123)
for(i in 1:B)
  estB[i] <- mean(sample(rain,size=n,replace = TRUE))

muEstB <- round(mean(estB),2)
muEstB

## [1] 9.28

sdEstB <- round(sd(estB),2)
sdEstB
```

```
## [1] 0.55
# MLE beta
estimate <- round(mean(rain),2)
estimate

## [1] 9.29
# Bias
bias <- round(muEstB-estimate,3)
bias

## [1] -0.01
# Bias corrected estimate
bcorrest <- estimate-bias
bcorrest

## [1] 9.3
# MSE
bMSE <- round(sdEstB^2 + bias^2,3)
bMSE

## [1] 0.303
rbind('beta estimate', estimate, sdEstB, bias, bcorrest, bMSE)

##           [,1]
##           "beta estimate"
## estimate "9.29"
## sdEstB   "0.55"
## bias     "-0.01"
## bcorrest "9.3"
## bMSE     "0.303"

ii)
# Bootstrap confidence intervals

#Normal
lowUpBound <- qnorm(0.975)*sdEstB
lowerN <- round(estimate-lowUpBound,2)
upperN <- round(estimate+lowUpBound,2)

# Basic
lowerB <- round(2*estimate-quantile(estB,0.975),2)
upperB <- round(2*estimate-quantile(estB,0.025),2)

# Percentile
lowerCI <- round(quantile(estB,0.025),2)
upperCI <- round(quantile(estB,0.975),2)

intervals <- c("Normal L", "Normal U", "Basic L", "Basic U",
              "Perc. L", "Perc. U")
rbind(intervals, c(lowerN, upperN, lowerB, upperB, lowerCI, upperCI))
```

##			97.5%	2.5%	2.5%	97.5%
## intervals	"Normal L"	"Normal U"	"Basic L"	"Basic U"	"Perc. L"	"Perc. U"
##	"8.21"	"10.37"	"8.22"	"10.4"	"8.18"	"10.36"

e)

Assume storms follow a homogeneous Poisson process with rate $\lambda = 6$ storms/month. Let $N(t)$ be the number of storms in t months.

Questions: i) What are the expected number of storms and the variance of the number of storms during any t -month period? ii) We are interested to estimate the probability that Stavanger experiences 90 or more storms in one year. Describe how we could simulate the numbers of storm in one year, and how we could estimate the probability to have 90 or more storms in one year.

i) As we assume the storms follow a homogeneous Poisson process with rate, we have that $N(t)$ is Poisson distributed with expectation $\lambda \cdot t$, and thus $Var(N(t)) = \lambda \cdot t$.

ii) We have to simulate a homogenous Poisson process with intensity λ .

Option 1:

Simulate interarrival times of storms $T_1, T_2, \dots$ from the exponential distribution with parameter $\lambda = 6$. Stop simulating interarrival times if $\sum_{i=1}^k T_i \leq 12$ and $\sum_{i=1}^{n+1} T_i > 12$ and keeping the interarrival times $T_1, \dots, T_k$.

Option 2:

First simulate the number of storms, k , in an interval $[0, 12]$ from the Poisson distribution with expectation $\lambda t = 6 \cdot 12 = 72$. Then simulate the arrival times inside this interval by simulating k random numbers from the Uniform $[0, 12]$ distribution and sort the numbers.

To estimate the probability to have 90 or more storms in one year, we can simulate e.g. $M = 5000$ Poisson processes with $N_m(12)$, $m = 1, \dots, M$ and calculated the probability by a frequency estimator

$$P(N(12) \geq 90) = \frac{\#(N_m(12) \geq 90)}{M}.$$

f)

Let us assume we want to model the number of storms as a non-homogenous poisson process. For this we consider a intensity $\lambda(t)$ as shown below. i) What is the interpretation of the density function? ii) Does homogenous or non-homogenous poisson process fits better to model the number of storms? Reason your answer! iii) Describe how to model a non-homogenous poisson process given the intensity $\lambda(t)$ as shown below.

i) The depicted intensity function suggests a change in storm intensity throughout the year, with more storms occurring during the winter (peaking around December, $\lambda(12) = 10$), and a dip around June ($\lambda(6) = 2$), where the intensity is lowest.

ii) A non-homogeneous Poisson process provides a better fit for modeling the number of storms, as we expect seasonal variation in Norway in the number of storms per month. It is more likely to observe more storms during the fall and winter than during the spring and summer.

iii) From the figure we see that $\lambda(t)$ is bounded with $\lambda(t) \leq \lambda_{max} = 10 \forall t \in [0, 12]$. Then we can use the "thinning method" to simulate a non-homogeneous Poisson process with intensity $\lambda(t)$ by

1. Simulate an HPP (as described above)) with intensity $\lambda_{max} = 10$, giving simulated arrival times $s_1, s_2, \dots, s_N$ of storms.
2. Simulate N variables $u_1, \dots, u_N$ from the $U[0, 1]$ distribution.
3. For each simulated arrival time s_i , accept the time with probability $\lambda(s_i)/10$, i.e. accept the time if $u_i \leq \lambda(s_i)/10$, otherwise delete this time.
4. Return the remaining arrival times.

g)

Suppose historical weather records for Stavanger suggest that the average rainfall per storm is typically around $8mm$, but with some uncertainty. We want to update this belief using the new observed rainfall data (the 40 depths from the dataset above). Assume a prior distribution $\beta \sim \text{Gamma}(a_0 = 4, b_0 = 2)$ with prior mean of $8mm$.

Show that the log-posterior density kernel is given by

$$\log(\pi(\beta|x)) \propto \log(\beta) = (a_0 - n - 1) \log(\beta) - \frac{\beta}{b_0} - \frac{\sum x_i}{\beta}.$$

We have for the posterior $\pi(\beta|x) \propto L(\beta) \cdot p(\beta)$ with Likelihood $L(\beta) = \beta^{-n} e^{-\frac{1}{\beta} \sum_{i=1}^n x_i}$ (see 1.b.i) and prior $p(\beta) \propto \beta^{a_0-1} e^{-\beta/b_0}$ which is a Gamma-density kernel. Thus

$$\begin{aligned} \pi(\beta|x) &\propto L(\beta) \cdot p(\beta) = \beta^{a_0-1} \beta^{-n} e^{-\beta/b_0} e^{-\frac{1}{\beta} \sum_{i=1}^n x_i} \\ &= \beta^{a_0-n-1} e^{-\beta/b_0 - \frac{1}{\beta} \sum_{i=1}^n x_i} \\ \log(\pi(\beta|x)) &\propto (a_0 - n - 1) \log(\beta) - \beta/b_0 - \frac{1}{\beta} \sum_{i=1}^n x_i. \end{aligned}$$

h)

Below we see R-code with an implementation of a MCMC algorithm. What kind of MCMC algorithm is this, and how is the performance of the algorithm given the R-output? Why is the below implementation not suitable for the target distribution? Reason your answers!

This is an Random Walk Metropolis Hasting (RWMH) algorithm. Based on the R- output the RWMH algorithm seems to perform well. The acceptance rate is 37% (i.e. between 20% – 40%), and the effective sample size is good with 4088 out of 20000 samples. So, the algorithm seems to be well tuned for the underlying task. Also the MCMC chain looks fine. However, the simple RWMH assumes for the proposal mechanism $\beta^* = \beta_{t-1} + \varepsilon$, $\varepsilon \sim N(0, \sigma^2)$ which can produce negative proposal for β , although $\beta > 0$. We could try to fix this by rejecting negatives (i.e., we set acceptance probability to 0 if $\beta^* \leq 0$) but this would breaks the symmetry, and could make the mixing poor near the boundary. Thus, it is better to work with a MH algorithm, where we use a proposal with only positive values. The RWMH works still fine as the target distribution is between 5 and 17, and we use $\sigma = 4$ for the normal proposal.

i)

Now we want to improve the MCMC algorithm for estimating the posterior.

Suggest a log-scale Metropolis-Hasting proposal with $\log(\beta^*) = \log(\beta_{t-1}) + \varepsilon$ and $\varepsilon \sim N(0, \sigma^2)$. Find the proposal density $q(\cdot|\beta_{t-1})$ for the proposal mechanism, and describe how to compute the acceptance probability $\log(\alpha)$.

The two suggested proposal here, suggest only positive values. The first one is log-scale Metropolis-Hasting proposal with $\log(\beta^*) = \log(\beta_{t-1}) + \varepsilon$ and $\varepsilon \sim N(0, 1)$.

$$\log(\beta^*) = \log(\beta_{t-1}) + \varepsilon \Leftrightarrow e^{\log(\beta^*)} = e^{\log(\beta_{t-1}) + \varepsilon}$$

Thus, if $\varepsilon \sim N(0, \sigma^2)$, then $\log(\beta_{t-1}) + \varepsilon \sim N(\log(\beta_{t-1}), \sigma^2)$, and

$$e^{\log(\beta_{t-1}) + \varepsilon} \sim \text{Lognormal}(\log(\beta_{t-1}), \sigma^2)$$

Thus, the proposal density $q(\cdot|\beta_{t-1}) \sim \text{Lognormal}(\log(\beta_{t-1}), \sigma^2)$ is the pdf of a log-normal distribuion. Further, if we assume the log target density kernel

$$\log(g(\beta)) = (a_0 - n - 1) \log(\beta) - \beta/b_0 - \frac{1}{\beta} \sum_{i=1}^n x_i$$

then the accept probability is

$$\log(\alpha) = \min \left(0, \log(g(\beta^*)) + \log(q(\beta_{t-1}|\beta^*)) - \left(\log(g(\beta_{t-1})) + \log(q(\beta^*|\beta_{t-1})) \right) \right).$$

j)

Suggest an independent Gamma proposal distribuion that is centred near the MLE $\hat{\beta}$, i.e. with mean $\hat{\beta}$ and variance near the estimated sample variance of $\hat{\beta}$. Find the proposal density $q(\cdot)$ including parameters, and describe how to compute the acceptance probability $\log(\alpha)$.

We construct a independent Gamma proposal centered near the MLE $\hat{\beta} = \bar{X}$, i.e. with mean near $\hat{\beta}$ and variance near the estimated sample variance of $\hat{\beta}$. Let us assume, we propose values $\beta^* \sim \text{Gamma}(a_p, b_p)$ where a_p is the shape and b_p the scale parameter in the Gamma distribution. Further, from 1b), we know that $E(\hat{\beta}) = E(\bar{X}) = \beta$ and $\text{Var}(\hat{\beta}) = \text{Var}(\bar{X}) = \frac{\beta^2}{n}$.

Thus, we have the two equations $a_p b_p = \hat{\beta}$ (proposal centred near MLE) and $a_p b_p^2 = \frac{\hat{\beta}^2}{n}$ (proposal variance near estimated sample variance). Plugging the first equation into the second gives us, $a_p b_p^2 = \frac{(a_p b_p)^2}{n}$, i.e., $a_p = n$, and thus we get for the first equation $n b_p = \hat{\beta}$, i.e. $b_p = \frac{1}{n} \hat{\beta}$. Thus, the proposal mechanism is

$$q(\beta^*) \sim \text{Gamma}(n, \hat{\beta}/n)$$

and

$$\log(\alpha) = \min \left(0, \log(g(\beta^*)) - \log(q(\beta^*)) - \left(\log(g(\beta_{t-1})) - \log(q(\beta_{t-1})) \right) \right)$$

or

$$\log(\alpha) = \min \left(0, \log(g(\beta^*)) + \log(q(\beta_{t-1})) - \left(\log(g(\beta_{t-1})) + \log(q(\beta^*)) \right) \right).$$

Problem 2

Assume we want to estimate the integral

$$I = \int_0^\infty \ln(1 + 4x^2) e^{-\frac{1}{2}x} dx.$$

a)

Write down an expression for a Monte Carlo estimator of I based on n random draws from an exponential distribution. In addition, describe the steps involved in constructing and using this estimator.

Write

$$I = \int_0^\infty \ln(1 + 4x^2) e^{-\frac{1}{2}x} dx = \int_0^\infty 2 \ln(1 + 4x^2) f(x) dx$$

where $f(x) = \frac{1}{2}e^{-\frac{1}{2}x}$ is the density function of an exponential distribution with parameter $\beta = 2$. Then, a Monte Carlo estimator is given by

$$\hat{I} = \frac{1}{n} \sum_{i=1}^n 2 \ln(1 + 4X_i^2)$$

where $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(\beta = 2)$.

b)

Show that the inverse cumulative distribution function of the Exponential distribution with scale parameter β is given as $F_X^{-1}(U) = -\beta \ln(U)$.

We have

$$F(x) = \int_0^x \frac{1}{\beta} e^{-\frac{1}{\beta}y} dy = [-e^{-\frac{1}{\beta}y}]_0^x = -e^{-\frac{1}{\beta}x} - (-1) = 1 - e^{-\frac{1}{\beta}x}$$

.

Further,

$$F(X) = U \Leftrightarrow 1 - e^{-\frac{1}{\beta}X} = U \Leftrightarrow e^{-\frac{1}{\beta}X} = 1 - U \Leftrightarrow -\frac{1}{\beta}X = \log(1 - U) \Leftrightarrow X = -\beta \log(1 - U) = -\beta \log(U).$$

Thus, $F_X^{-1}(U) = -\beta \log(U)$.

c)

Write down an expression for an antithetic Monte Carlo estimator of I based on n random draws. In addition, describe the steps involved in constructing and using this estimator. Do you think the antithetic Monte Carlo estimator works well for I ? Why?

For the antithetic Monte Carlo estimator, we need to create negatively correlated exponential distributed variables. This can be achieved by $U_i \sim U(0, 1)$ and $V_i \sim 1 - U_i$, $i = 1, \dots, n/2$, and plugging these antithetic variables into the inverse cumulative density function, i.e., $X_i = F^{-1}(U_i) = -\beta \log(U_i)$ and $Y_i = F^{-1}(V_i)$.

Finally, an antithetic Monte Carlo estimator of I based on n random draws is given by

$$\hat{I}_{AT} = \frac{1}{n} \sum_{i=1}^{n/2} h(X_i) + h(Y_i)$$

with $h(x) = 2 \ln(1 + 4x^2)$.

The antithetic Monte Carlo estimator should improve our estimate, because $h(x) = 2 \ln(1 + 4x^2)$, and $1 + 4x^2 \geq 0$ for $x \geq 0$, and the logarithm function is a monotonic function. Thus $h(x)$ is a monotonic function and it is ensured that the antithetic variables will improve the estimate.